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Garden of Knowledge and Virtue

COURSE: DEEP LEARNING

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**Course Project Report: Geospatial Building Detection from Satellite
Imagery**

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ABSTRACT

This project presents a comparative performance evaluation of three prominent deep learning architectures, YOLOv8, YOLOv11, and Faster R-CNN (ResNet50), applied to the task of building footprint detection in high-altitude satellite imagery. Utilizing a highly constrained dataset of 55 annotated satellite patches, the study analyzes the trade-offs between architectural capacity, data augmentation strategies, and generalization capability. Empirical results demonstrate that while two-stage architectures suffer severely from data starvation and next-generation attention-heavy models overfit to localized noise, YOLOv8 emerges as the most resilient and optimal model for data-constrained geospatial tasks. Additionally, a robust post-processing pipeline was engineered to ensure strict structural compliance with Kaggle leaderboard ingestion formatting.

INTRODUCTION AND PROBLEM STATEMENT

Automated building extraction from remote sensing data plays a critical role in urban planning, disaster response, and geospatial intelligence. However, utilizing computer vision for satellite imagery presents unique challenges:

- **Scale and Orientation Variance:** Buildings exhibit highly diverse structural geometries, varying roof textures, and complex shadows.
- **Data Scarcity:** Manual annotation of high-resolution geospatial imagery is labor-intensive and expensive, frequently leaving practitioners with small, data-constrained training sets.

This project addresses these challenges by benchmarking state-of-the-art one-stage object detectors against a classic two-stage framework to determine the most effective architecture under strict data limitations (55 raw image patches).

BACKGROUND CONCEPT

To balance detection accuracy and computational overhead, the project evaluates two distinct paradigms in object detection:

1. One-Stage Detectors (YOLO Family)

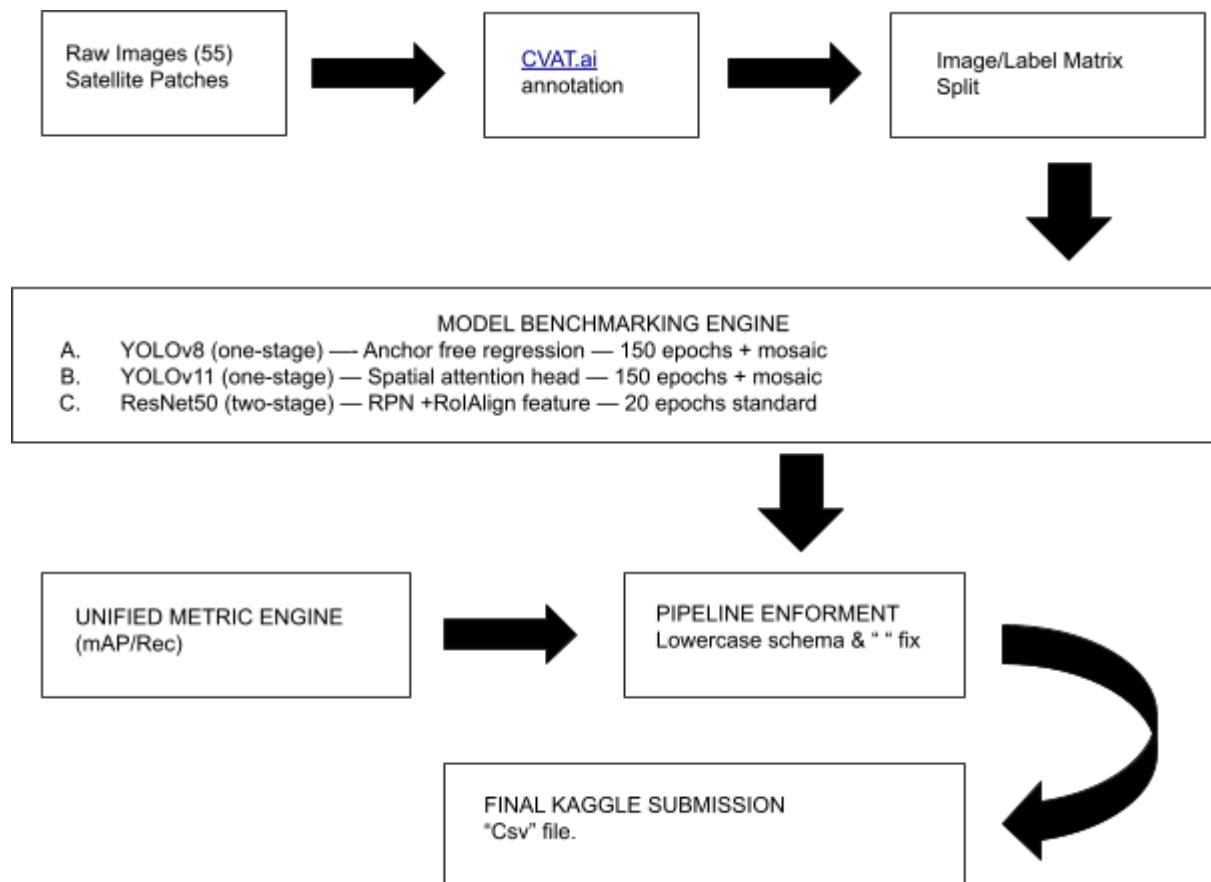
One-stage detectors treat object detection as a single regression problem, mapping pixels directly to bounding box coordinates and class probabilities.

- **YOLOv8:** It utilizes an anchor-free detection head, decoupling the classification and bounding-box regression tasks. This reduces structural rigidity and improves convergence speed.
- **YOLOv11:** Incorporates advanced multi-scale feature extraction layers and spatial attention heads designed to capture complex long-range dependencies across the image matrix.

2. Two-Stage Detectors (Faster R-CNN)

- **Faster R-CNN (ResNet50):** It divides the task into two distinct phases. First, a Region Proposal Network (RPN) identifies candidate Regions of Interest (RoIs). Second, RoIAlign extracts features from these proposals, passing them to a ResNet50 backbone for final classification and bounding box refinement.

METHODOLOGY AND SYSTEM ARCHITECTURE



DATA ENGINEERING AND AUGMENTATION

- **Annotation:** Raw high-altitude satellite patches were manually annotated using CVAT.ai, generating precise bounding boxes suitable for YOLO-formatted matrices.
- **The Mosaic Augmentation Factor:** To mitigate the severe data constraints of the 55-image distribution, the YOLO pipelines integrated Mosaic Augmentation. This technique combines four training images into a single composite frame, forcing the model to detect objects at smaller scales and significantly increasing localized background variety.

RESULTS AND ANALYSIS

Architecture Model	mAP @ 0.5	mAP @ 0.5:0.95	Precision @ 0.5	Recall
YOLOv8	0.6949	0.3507	0.4164	0.5142
YOLOv11	0.6524	0.2972	0.3873	0.4565
ResNet50	0.2860	0.0891	0.1667	0.3261

DISCUSSION

1. Why the YOLO Family Outperformed Faster R-CNN?
 - The two-stage Faster R-CNN architecture scored significantly lower (mAP = 0.2860) due to data starvation. Lacking native integration with advanced, synthetic data augmentations like Mosaic, its deep ResNet50 backbone could not generalize effectively on the raw 55-image distribution. Conversely, the YOLO models dynamically multiplied their data variety during training, minimizing overfitting and learning robust structural variations.
2. The Overfitting Phenomenon: YOLOv8 vs. YOLOv11.
 - YOLOv8 marginally outscored its newer counterpart, YOLOv11. While YOLOv11 possesses a higher model capacity and complex multi-scale attention layers, it suffers from subtle overfitting given the severe constraints of the small dataset. Instead of mapping structural generalities of building footprints, YOLOv11 began tracking localized environmental noise.

CONCLUSION AND FUTURE WORK

While YOLOv11 represents the current cutting edge of the YOLO lineage, YOLOv8 proved to be the most robust, resilient, and optimal architecture for data-constrained geospatial tasks. It stands as the definitive champion of this comparative study and serves as the primary candidate deployment for the Kaggle private leaderboard.

FUTURE RECOMMENDATIONS

1. **Expand Dataset Scale:** Integrate automated synthetic data generation or semi-supervised labeling pipelines to expand the base asset size beyond 55 patches.
2. **Cross-Domain Fine-Tuning:** Pre-train the architectural backbones on larger public geospatial datasets before fine-tuning on localized features.